Work Trail — Data Model

Use PostgreSQL. Enable PostGIS if map/geospatial querying is needed at scale. Store coordinate values as numeric columns initially; add a geography point/index when implementing proximity/geofence features.

Entity relationships

```
Company 1--* User
Company 1--* Employee
User 0..1--1 Employee
Employee 1--* WorkSession
WorkSession 1--* LocationPoint
WorkSession 1--* WorkUpdate
Employee 1--* GeneratedSummary
Company 1--* Report
```

Tables

`companies`

Column	Type	Notes
id	UUID PK	
name	varchar(160)	
timezone	varchar(64)	default `Asia/Kolkata`
precise_location_retention_days	integer	default 90
created_at, updated_at	timestamptz	

`users`

Column	Type	Notes
id	UUID PK	
company_id	UUID FK	indexed
role	enum	`OWNER`, `ADMIN`, `EMPLOYEE`
name	varchar(160)	
email	citext	unique per company or globally based on auth policy
phone	varchar(24)	
password_hash	text	never expose

Column	Type	Notes
status	enum	`INVITED`, `ACTIVE`, `DISABLED`
created_at, updated_at	timestamptz	

`employees`

Column	Type	Notes
id	UUID PK	
company_id	UUID FK	indexed
user_id	UUID FK nullable	populated after invite acceptance
employee_code	varchar(80)	unique within company
designation	varchar(120)	
created_at, updated_at	timestamptz	

`invitations`

Column	Type	Notes
id	UUID PK	
company_id, employee_id	UUID FK	
token_hash	text	store hash, never plaintext token
expires_at	timestamptz	
accepted_at	timestamptz nullable	
created_by_user_id	UUID FK	

`work_sessions`

Column	Type	Notes
id	UUID PK	
company_id, employee_id	UUID FK	indexed
status	enum	`ACTIVE`, `COMPLETED`, `INCOMPLETE`
started_at_server, started_at_device	timestamptz	
ended_at_server, ended_at_device	timestamptz nullable	
total_minutes	integer nullable	derived after completion
created_at, updated_at	timestamptz	

Add a partial unique index: one `ACTIVE` session per employee.

`location_points`

Column	Type	Notes
id	UUID PK	
company_id, employee_id, session_id	UUID FK	indexed
client_point_id	UUID	unique with employee for retry dedupe
latitude, longitude	numeric(10,7)	
accuracy_meters	numeric(8,2)	
speed_mps, bearing	numeric nullable	
battery_level	numeric(4,3) nullable	0–1
source	enum	`START`, `BACKGROUND`, `END`, `MANUAL`
captured_at	timestamptz	device capture time
received_at	timestamptz	server time
quality_flag	enum	`GOOD`, `LOW_ACCURACY`, `SUSPICIOUS`, `INVALID`

Indexes: `(employee_id, captured_at)`, `(session_id, captured_at)`, and organisation/time index for retention deletion.

`work_updates`

Column	Type	Notes
id	UUID PK	
company_id, employee_id, session_id	UUID FK	indexed
text	text	final editable employee text
input_method	enum	`TYPED`, `VOICE`
category	varchar(64) nullable	
submitted_at	timestamptz	
summary_status	enum	`QUEUED`, `PROCESSING`, `COMPLETE`, `FAILED`
ai_summary	text nullable	
ai_structured_data	jsonb nullable	validated schema
model_name	varchar(120) nullable	

`generated_summaries`

Column	Type	Notes
id	UUID PK	
company_id, employee_id	UUID FK	indexed

Column	Type	Notes
period_type	enum	`DAILY`, `WEEKLY`, `MONTHLY`
period_start, period_end	timestamptz	
source_update_count	integer	
source_session_minutes	integer	
content	text	AI generated
structured_data	jsonb	validated output
status	enum	`QUEUED`, `COMPLETE`, `FAILED`
generated_at	timestamptz nullable	

Unique index: `(employee_id, period_type, period_start, period_end)`.

`reports`

Column	Type	Notes
id	UUID PK	
company_id, employee_id	UUID FK	
requested_by_user_id	UUID FK	
period_start, period_end	timestamptz	
status	enum	`QUEUED`, `GENERATING`, `READY`, `FAILED`
storage_key	text nullable	private bucket path
generated_at, expires_at	timestamptz nullable	

`audit_logs`

Store actor, company, action, entity type/id, IP/device metadata where appropriate, and timestamp. Audit at minimum invitation creation/acceptance, report export, role changes, session overrides, and deletion requests.